

Phase 3: FDM, Images, & Cables

Advanced Boundary Value Problems

Master Study Guide & Step-by-Step Derivations

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1 The Uniqueness Theorem

Before we can use any advanced analytical "tricks" to solve complex electrostatic fields, we must mathematically prove that our tricks are valid. This is the sole purpose of the Uniqueness Theorem.

1.1 The Statement of the Theorem

Statement:

If a solution to Laplace's equation ($\nabla^2 V = 0$) or Poisson's equation ($\nabla^2 V = -\rho_v/\epsilon$) satisfies a given set of fixed boundary conditions over a closed boundary surface S defining a specified internal volume τ , then it is the *one and only* (unique) physical solution possible for that entire region.

Why this matters:

If you find a potential function $V(x, y, z)$ by completely guessing, or by pretending the physical setup is different (like in the Method of Images), but your guessed function perfectly matches the required voltages on the boundaries, you are guaranteed that your guess is the 100% correct reality.

1.2 Rigorous Proof (Highly Tested PYQ)

We will prove this using the method of contradiction.

Step 1: The Assumption

Assume that the solution is *not* unique. Suppose there exist two completely different scalar potential functions, V_1 and V_2 , that both satisfy Poisson's equation within the exact same volume τ .

$$\nabla^2 V_1 = -\frac{\rho_v}{\epsilon} \quad \text{inside } \tau \quad (1)$$

$$\nabla^2 V_2 = -\frac{\rho_v}{\epsilon} \quad \text{inside } \tau \quad (2)$$

Because both purported solutions claim to be valid, they must both perfectly match the known, physical boundary voltages on the outer surface S :

$$V_1 = V_2 \quad \text{on boundary surface } S \quad (3)$$

Step 2: The Difference Potential

Let us define a new artificial mathematical function, the "difference potential", denoted V_d :

$$V_d = V_1 - V_2 \quad (4)$$

What are the properties of this difference potential? Inside the volume τ , we subtract Equation 2 from Equation 1:

$$\nabla^2 V_d = \nabla^2 V_1 - \nabla^2 V_2 \quad (5)$$

$$\nabla^2 V_d = \left(-\frac{\rho_v}{\epsilon}\right) - \left(-\frac{\rho_v}{\epsilon}\right) \quad (6)$$

$$\nabla^2 V_d = 0 \quad (7)$$

Thus, V_d always perfectly satisfies Laplace's equation.

On the boundary S , because $V_1 = V_2$, their difference is zero:

$$V_d = 0 \quad \text{on boundary } S \quad (8)$$

Step 3: Vector Calculus Identity

We employ a standard vector identity for the divergence of a scalar (f) multiplied by a vector field ($\mathbf{A}$):

$$\nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f) \quad (9)$$

Let the scalar be $f = V_d$, and the vector field be its own gradient $\mathbf{A} = \nabla V_d$:

$$\nabla \cdot (\mathbf{V}_d \nabla V_d) = V_d(\nabla \cdot (\nabla V_d)) + (\nabla V_d) \cdot (\nabla V_d) \quad (10)$$

Recall two facts: the divergence of a gradient is the Laplacian (∇^2), and the dot product of a vector with itself is its magnitude squared.

$$\nabla \cdot (\mathbf{V}_d \nabla V_d) = V_d(\nabla^2 V_d) + |\nabla V_d|^2 \quad (11)$$

Since we already proved $\nabla^2 V_d = 0$ inside the volume, the first term on the right vanishes completely:

$$\nabla \cdot (\mathbf{V}_d \nabla V_d) = |\nabla V_d|^2 \quad (12)$$

Step 4: Integration and the Divergence Theorem

Integrate this identity over the entire volume τ :

$$\iiint_{\tau} \nabla \cdot (\mathbf{V}_d \nabla V_d) d\tau = \iiint_{\tau} |\nabla V_d|^2 d\tau \quad (13)$$

Apply the Divergence Theorem (Gauss's Theorem) to the left side to convert the volume integral of a divergence into a closed surface integral:

$$\oint_S (V_d \nabla V_d) \cdot d\mathbf{S} = \iiint_{\tau} |\nabla V_d|^2 d\tau \quad (14)$$

Step 5: The Collapse

Recall Equation 8: $V_d = 0$ perfectly on the boundary surface S . Because V_d is a scalar multiplier in the surface integral, the entire left side of the equation evaluates to exactly zero.

$$0 = \iiint_{\tau} |\nabla V_d|^2 d\tau \quad (15)$$

Step 6: Logical Resolution

The term $|\nabla V_d|^2$ is the square of a real magnitude. It must be strictly non-negative (greater than or equal to zero). If you integrate a strictly non-negative function over a volume and the result is zero, the function itself must be zero everywhere.

$$|\nabla V_d|^2 = 0 \implies \nabla V_d = 0 \quad (16)$$

If the gradient of a scalar is zero everywhere, the scalar must be a constant:

$$V_d = C \quad (17)$$

But since we know $V_d = 0$ on the boundary, and potential must be continuous, the constant C must be zero.

$$V_d = 0 \quad (18)$$

$$V_1 - V_2 = 0 \quad (19)$$

$$V_1 = V_2 \quad (20)$$

Our assumption that two different solutions could exist is mathematically false. The solution is unique. (Q.E.D.)

2 The Method of Images

Now that we have the Uniqueness Theorem, we can use it to "cheat".

If a positive point charge $+Q$ is placed above an infinite, flat, grounded metal plate, the positive charge pulls electrons from the earth, creating a non-uniform puddle of negative charge on the plate. Calculating the field by integrating over this puddle is practically impossible.

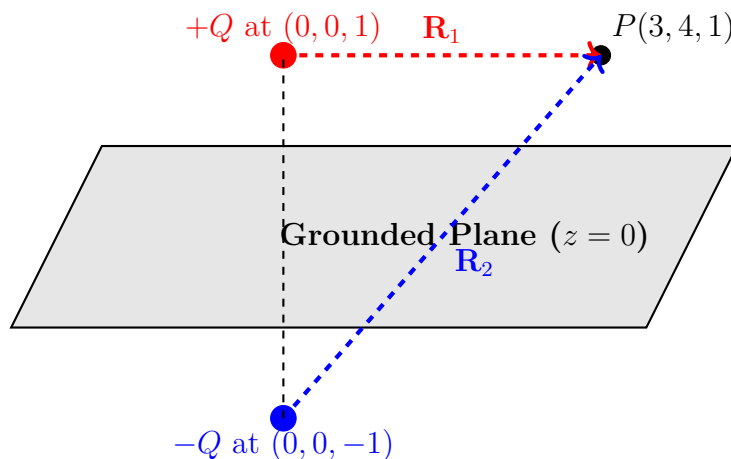
However, the grounded plate imposes a simple boundary condition: $V = 0$ everywhere on the $z = 0$ plane. If we remove the metal plate entirely, and place a fictitious "Image Charge" $-Q$ exactly symmetrically below the boundary, the superposition of $+Q$ and $-Q$ naturally creates a $V = 0$ plane exactly where the metal plate used to be.

By the Uniqueness Theorem, the electric field calculated in the upper region ($z > 0$) using this fake two-charge system is **identically equal** to the real electric field produced by the complex metal plate.

2.1 Point Charge Stresses (PYQ 2022)

The Problem: A point charge of 1 mC is placed at $(0, 0, 1)$ m. An infinitely long conducting plane is there at $z = 0$. Find the electric potential and the electric stresses at the point $(3, 4, 1)$ m. Medium is air.

TikZ Diagram: Method of Images Setup



Step 1: Identify Parameters and Vectors

Real Charge: $Q = 10^{-3}$ C at $\mathbf{r}_A = (0, 0, 1)$.

Image Charge: $-Q = -10^{-3}$ C at $\mathbf{r}_B = (0, 0, -1)$.

Observation Point: P at $\mathbf{r}_P = (3, 4, 1)$.

Distance vector from Real Charge to P:

$$\mathbf{R}_1 = \mathbf{r}_P - \mathbf{r}_A = (3 - 0)\hat{\mathbf{i}} + (4 - 0)\hat{\mathbf{j}} + (1 - 1)\hat{\mathbf{k}} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}} \quad (21)$$

Magnitude $R_1 = \sqrt{3^2 + 4^2} = 5$ meters.

Distance vector from Image Charge to P:

$$\mathbf{R}_2 = \mathbf{r}_P - \mathbf{r}_B = (3 - 0)\hat{\mathbf{i}} + (4 - 0)\hat{\mathbf{j}} + (1 - (-1))\hat{\mathbf{k}} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + 2\hat{\mathbf{k}} \quad (22)$$

Magnitude $R_2 = \sqrt{3^2 + 4^2 + 2^2} = \sqrt{9 + 16 + 4} = \sqrt{29}$ meters.

Step 2: Calculate Electric Potential (V)

By superposition, $V_P = V_{\text{real}} + V_{\text{image}}$:

$$V_P = \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{R_1} + \frac{-Q}{R_2} \right) \quad (23)$$

$$V_P = (9 \times 10^9) \times 10^{-3} \left(\frac{1}{5} - \frac{1}{\sqrt{29}} \right) \quad (24)$$

$$V_P = 9 \times 10^6 (0.2 - 0.1857) \quad (25)$$

$$V_P = 9 \times 10^6 \times 0.0143 = 128.7 \text{ kV} \quad (26)$$

Step 3: Calculate Electric Stresses

Electric Stress is identical to Energy Density ($w_e = \frac{1}{2}\epsilon_0|\mathbf{E}|^2$). We must find the net vector field $\mathbf{E}_{\text{net}}$ first.

Field from Real Charge:

$$\mathbf{E}_1 = \frac{Q}{4\pi\epsilon_0 R_1^3} \mathbf{R}_1 = \frac{9 \times 10^6}{5^3} (3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}) = 216\hat{\mathbf{i}} + 288\hat{\mathbf{j}} \text{ kV/m} \quad (27)$$

Field from Image Charge:

$$\mathbf{E}_2 = \frac{-Q}{4\pi\epsilon_0 R_2^3} \mathbf{R}_2 = \frac{-9 \times 10^6}{(\sqrt{29})^3} (3\hat{\mathbf{i}} + 4\hat{\mathbf{j}} + 2\hat{\mathbf{k}}) = -172.9\hat{\mathbf{i}} - 230.5\hat{\mathbf{j}} - 115.3\hat{\mathbf{k}} \text{ kV/m} \quad (28)$$

Net Field $\mathbf{E}_{\text{net}} = \mathbf{E}_1 + \mathbf{E}_2$:

$$\mathbf{E}_{\text{net}} = 43.1\hat{\mathbf{i}} + 57.5\hat{\mathbf{j}} - 115.3\hat{\mathbf{k}} \text{ kV/m} \quad (29)$$

Calculate the magnitude squared (convert back to V/m for final formula):

$$|\mathbf{E}_{\text{net}}|^2 = (43.1)^2 + (57.5)^2 + (-115.3)^2 \text{ (kV/m)}^2 \quad (30)$$

$$|\mathbf{E}_{\text{net}}|^2 \approx 18450 \times 10^6 \text{ (V/m)}^2 = 1.845 \times 10^{10} \text{ (V/m)}^2 \quad (31)$$

Finally, calculate Stress:

$$\text{Stress} = \frac{1}{2}\epsilon_0|\mathbf{E}_{\text{net}}|^2 \quad (32)$$

$$\text{Stress} = 0.5 \times (8.854 \times 10^{-12}) \times (1.845 \times 10^{10}) \quad (33)$$

$$\text{Stress} = 0.0817 \text{ N/m}^2 \quad (34)$$

2.2 Line Charge Equipotentials (Proof of Cylinders)

The Problem: Prove that the equipotential surfaces generated by an infinitely long line charge and its image (with respect to an infinite conducting plane) are cylinders.

Step 1: Setup and Geometry

Assume the grounded conducting plane lies at $x = 0$. An infinitely long line charge $+\lambda$ (C/m) is placed at $x = a$. By the Method of Images, we replace the plane with an image line charge $-\lambda$ located symmetrically at $x = -a$.

Let $P(x, y)$ be any arbitrary observation point in the 2D cross-sectional plane. The distance from $+\lambda$ to P is $R_1 = \sqrt{(x - a)^2 + y^2}$. The distance from $-\lambda$ to P is $R_2 = \sqrt{(x + a)^2 + y^2}$.

Step 2: Formulate the Potential

The scalar potential of a single infinite line charge is $V = -\frac{\lambda}{2\pi\epsilon_0} \ln(R)$. By superposition, the total potential at P is:

$$V_P = \left(-\frac{\lambda}{2\pi\epsilon_0} \ln(R_1) \right) + \left(\frac{\lambda}{2\pi\epsilon_0} \ln(R_2) \right) \quad (35)$$

$$V_P = \frac{\lambda}{2\pi\epsilon_0} [\ln(R_2) - \ln(R_1)] \quad (36)$$

$$V_P = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{R_2}{R_1}\right) \quad (37)$$

Step 3: Impose Equipotential Condition

For a surface to be an equipotential, V_P must equal a constant value, V_0 .

$$V_0 = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{R_2}{R_1}\right) \quad (38)$$

Rearranging to isolate the distance ratio:

$$\ln\left(\frac{R_2}{R_1}\right) = \frac{2\pi\epsilon_0 V_0}{\lambda} \quad (39)$$

Exponentiate both sides. Because the right side is entirely composed of constants, we define a new scalar constant K :

$$\frac{R_2}{R_1} = e^{\frac{2\pi\epsilon_0 V_0}{\lambda}} = K \quad (40)$$

Square both sides to remove the square roots from the distance formulas:

$$\frac{R_2^2}{R_1^2} = K^2 \implies R_2^2 = K^2 R_1^2 \quad (41)$$

Step 4: Algebraic Expansion

Substitute our Cartesian distance definitions into Equation 41:

$$(x + a)^2 + y^2 = K^2 [(x - a)^2 + y^2] \quad (42)$$

Expand the binomials carefully:

$$x^2 + 2ax + a^2 + y^2 = K^2(x^2 - 2ax + a^2 + y^2) \quad (43)$$

Group all the x^2 , x , y^2 , and constant terms together on the left side:

$$x^2(1 - K^2) + y^2(1 - K^2) + 2ax(1 + K^2) + a^2(1 - K^2) = 0 \quad (44)$$

Divide the entire equation by the common coefficient $(1 - K^2)$:

$$x^2 + y^2 + 2ax \left(\frac{1 + K^2}{1 - K^2} \right) + a^2 = 0 \quad (45)$$

Step 5: Completing the Square

Let us define a new grouped constant $M = a \left(\frac{1+K^2}{1-K^2} \right)$ to make the algebra cleaner.

$$x^2 + 2Mx + y^2 + a^2 = 0 \quad (46)$$

To complete the square for the x terms, we add and subtract M^2 :

$$(x^2 + 2Mx + M^2) - M^2 + y^2 + a^2 = 0 \quad (47)$$

$$(x + M)^2 + y^2 = M^2 - a^2 \quad (48)$$

Conclusion:

This final equation represents the standard geometric formula for a circle: $(x - h)^2 + (y - k)^2 = R_c^2$. The center of the circle lies on the x-axis at $h = -M$, and the radius squared is $R_c^2 = M^2 - a^2$. Because this is a 2D cross-section of physical line charges that extend infinitely along the z-axis (into and out of the page), these circular cross-sections physically sweep out 3D **cylinders**. The proof is complete.

3 The Finite Difference Method (FDM)

When boundaries form complex shapes (like the jagged corners of a microchip casing), analytical integration fails. We must instruct a computer to approximate the solution over a vast grid of nodes.

We know that in a charge-free region, the potential must satisfy Laplace's equation:

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0 \quad (49)$$

Our goal is to mathematically replace these continuous calculus derivatives with simple, discrete algebra (addition and subtraction of neighbor node voltages).

3.1 Derivation for Unequal Nodal Spacing

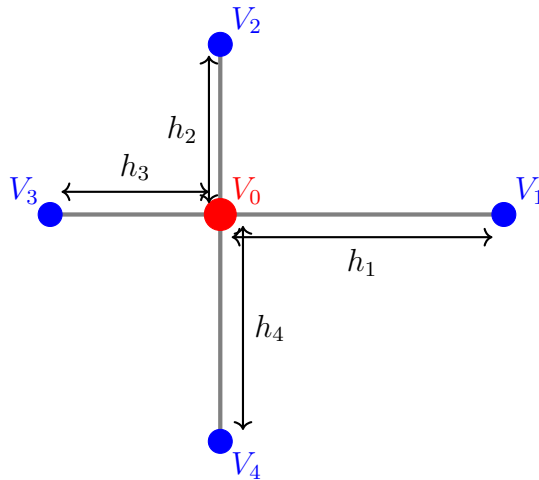
(Note: The equal spacing derivation is simply a special case of this broader proof).

Step 1: The Computational Stencil

Consider a central node 0 located at (x, y) with unknown potential V_0 . It is surrounded by four neighbors:

- Node 1 (Right) at distance h_1 with potential V_1 .
- Node 2 (Top) at distance h_2 with potential V_2 .
- Node 3 (Left) at distance h_3 with potential V_3 .
- Node 4 (Bottom) at distance h_4 with potential V_4 .

TikZ Diagram: Unequal FDM Stencil



Step 2: Taylor Series Expansions

We expand the potential function along the x-axis using the Taylor series.

Forward expansion (for V_1 at distance $+h_1$):

$$V_1 = V_0 + h_1 \frac{\partial V}{\partial x} + \frac{h_1^2}{2} \frac{\partial^2 V}{\partial x^2} + O(h_1^3) \quad (50)$$

Backward expansion (for V_3 at distance $-h_3$):

$$V_3 = V_0 - h_3 \frac{\partial V}{\partial x} + \frac{h_3^2}{2} \frac{\partial^2 V}{\partial x^2} + O(h_3^3) \quad (51)$$

Step 3: Eliminating the First Derivative

Our goal is to isolate the second derivative. We must strategically eliminate $\frac{\partial V}{\partial x}$. Multiply Equation 50 by h_3 , and Equation 51 by h_1 :

$$h_3 V_1 = h_3 V_0 + h_1 h_3 \frac{\partial V}{\partial x} + \frac{h_1^2 h_3}{2} \frac{\partial^2 V}{\partial x^2} \quad (52)$$

$$h_1 V_3 = h_1 V_0 - h_1 h_3 \frac{\partial V}{\partial x} + \frac{h_1 h_3^2}{2} \frac{\partial^2 V}{\partial x^2} \quad (53)$$

Add the two equations together. The first derivative terms perfectly annihilate each other:

$$h_3 V_1 + h_1 V_3 = (h_1 + h_3) V_0 + \left(\frac{h_1^2 h_3 + h_1 h_3^2}{2} \right) \frac{\partial^2 V}{\partial x^2} \quad (54)$$

Factor out the common term ($h_1 h_3$) from the bracket:

$$h_3 V_1 + h_1 V_3 = (h_1 + h_3) V_0 + \frac{h_1 h_3 (h_1 + h_3)}{2} \frac{\partial^2 V}{\partial x^2} \quad (55)$$

Step 4: Isolating the Second Derivative

Rearrange to solve for $\frac{\partial^2 V}{\partial x^2}$:

$$\frac{h_1 h_3 (h_1 + h_3)}{2} \frac{\partial^2 V}{\partial x^2} = h_3 V_1 + h_1 V_3 - (h_1 + h_3) V_0 \quad (56)$$

$$\frac{\partial^2 V}{\partial x^2} = \frac{2}{h_1 h_3 (h_1 + h_3)} [h_3 (V_1 - V_0) + h_1 (V_3 - V_0)] \quad (57)$$

Distributing the ($h_1 h_3$) denominator cleanly yields:

$$\frac{\partial^2 V}{\partial x^2} = \frac{2}{h_1 + h_3} \left[\frac{V_1 - V_0}{h_1} + \frac{V_3 - V_0}{h_3} \right] \quad (58)$$

By exact parallel logic along the y-axis:

$$\frac{\partial^2 V}{\partial y^2} = \frac{2}{h_2 + h_4} \left[\frac{V_2 - V_0}{h_2} + \frac{V_4 - V_0}{h_4} \right] \quad (59)$$

Substitute both into Laplace's equation to finish the derivation:

$$\frac{2}{h_1 + h_3} \left[\frac{V_1 - V_0}{h_1} + \frac{V_3 - V_0}{h_3} \right] + \frac{2}{h_2 + h_4} \left[\frac{V_2 - V_0}{h_2} + \frac{V_4 - V_0}{h_4} \right] = 0 \quad (60)$$

Special Case: Equal Spacing

If the grid is perfectly uniform ($h_1 = h_2 = h_3 = h_4 = h$), the equation collapses beautifully. All the h denominators factor out and cancel against the zero on the right side.

$$(V_1 - V_0) + (V_3 - V_0) + (V_2 - V_0) + (V_4 - V_0) = 0 \quad (61)$$

$$V_1 + V_2 + V_3 + V_4 - 4V_0 = 0 \quad (62)$$

$$V_0 = \frac{1}{4}(V_1 + V_2 + V_3 + V_4) \quad (63)$$

The potential at any point is simply the arithmetic average of its four neighbors!

4 Cables and Capacitance Grading

In a standard, un-graded coaxial cable (with a single dielectric), the electric field intensity at a radial distance x is derived from Gauss's Law:

$$E_x = \frac{q}{2\pi\epsilon x} \quad (64)$$

Because the electric field is inversely proportional to the radius ($1/x$), the absolute **maximum** stress naturally occurs exactly at the surface of the inner conductor (where $x = r$, the smallest radius). The field then decays logarithmically outwards.

This is highly inefficient. The inner dielectric is pushed to its breakdown limit, while the outer layers sit practically un-stressed.

4.1 The Field Utilization Factor (η)

Engineers measure this efficiency using the Field Utilization Factor. It is the ratio of the average field across the entire gap to the maximum peak field.

$$\eta = \frac{E_{\text{avg}}}{E_{\text{max}}} \quad (65)$$

For a standard coaxial cable (inner radius r , outer sheath R , voltage V):

$$E_{\text{avg}} = \frac{V}{R - r} \quad \text{and} \quad E_{\text{max}} = \frac{V}{r \ln(R/r)} \quad (66)$$

Substituting these into the definition:

$$\eta = \frac{\left(\frac{V}{R-r}\right)}{\left(\frac{V}{r \ln(R/r)}\right)} \quad (67)$$

$$\eta = \frac{r \ln(R/r)}{R - r} \quad (68)$$

4.2 Capacitance Grading

To achieve a more uniform stress distribution (pushing η closer to 1), engineers wrap the core in multiple concentric layers of dielectrics with different permittivities.

Suppose we have two dielectric layers:

- **Layer 1 (Inner):** Permittivity ϵ_1 , occupying $r_1 \leq x < r_2$.
- **Layer 2 (Outer):** Permittivity ϵ_2 , occupying $r_2 \leq x < R$.

The peak stress in *each* individual layer still occurs at its innermost boundary:

$$E_{\max 1} = \frac{q}{2\pi\epsilon_1 r_1} \quad (69)$$

$$E_{\max 2} = \frac{q}{2\pi\epsilon_2 r_2} \quad (70)$$

Ideal Grading vs. Reversed Grading:

If an engineer perfectly designs the cable, $E_{\max 1}$ will exactly equal $E_{\max 2}$. However, an exam question may ask for the mathematical condition where the global maximum stress does **NOT** occur at the innermost conductor surface (an "improper" or "reversed" grade).

This happens if Layer 2 is accidentally forced to contain the highest stress:

$$E_{\max 2} > E_{\max 1} \quad (71)$$

Substitute the formulas:

$$\frac{q}{2\pi\epsilon_2 r_2} > \frac{q}{2\pi\epsilon_1 r_1} \quad (72)$$

Cancel the common charge term q and constant 2π :

$$\frac{1}{\epsilon_2 r_2} > \frac{1}{\epsilon_1 r_1} \quad (73)$$

Cross-multiply the positive denominators to reach the final proven condition:

$$\epsilon_1 r_1 > \epsilon_2 r_2 \quad (74)$$

5 Phase 3 Summary and Exam Checklist

Congratulations! You have conquered the most mathematically intense portion of the syllabus. You can now prove the uniqueness of boundary value solutions, utilize the Method of Images to simplify complex physical geometries, deploy numerical FDM techniques for computers, and engineer graded high-voltage cables.

Before advancing to Phase 4 (Magnetostatics), ensure you can confidently check off every item on this list:

- ☐ I can write out the Uniqueness Theorem proof, explicitly understanding how the vector identity $\nabla \cdot (\mathbf{V}_d \nabla \mathbf{V}_d)$ mathematically forces the difference potential to be zero everywhere.
- ☐ I can fully expand the $(x+a)^2 + y^2 = K^2[(x-a)^2 + y^2]$ equation from the Method of Images line charge problem, complete the square, and prove it forms the equation of a geometric circle.
- ☐ I can derive the 5-point FDM stencil formula for unequal grid spacing using forward and backward Taylor Series expansions, carefully eliminating the first derivative.
- ☐ I can mathematically prove the condition required for a graded coaxial cable to experience its peak electric stress off the inner core ($\epsilon_1 r_1 > \epsilon_2 r_2$).

Once this checklist is complete, you are fully prepared to shift gears from stationary charges to flowing currents in **Phase 4: Magnetostatics**.